

Building a Reproducible Security Operations Centre Home Lab with an Open-Source SIEM

A Practical Study in Security Monitoring,
Endpoint Telemetry, and Alert Triage

Muhammad Ahmed Khalid

Personal Portfolio Project

This report documents a self-initiated technical project. It is not submitted for academic credit. It accompanies the project repository and is intended to demonstrate practical security-monitoring skills.

May 19, 2026

Abstract

Security Operations Centre (SOC) analysts depend on Security Information and Event Management (SIEM) platforms to detect and investigate threats, yet graduates entering the field rarely have hands-on experience with these tools. This project addresses that gap by designing, building, and documenting a fully functional SOC home lab using the open-source SIEM platform Wazuh.

The lab consists of a SIEM server and two monitored endpoints — a Windows host and a Linux host — deployed on an isolated virtual network. Enhanced endpoint telemetry is provided by Sysmon on Windows and the Linux Audit framework on Linux. The environment is used to generate security events, observe the resulting alerts, and demonstrate a realistic analyst triage workflow. A core aim of the project is reproducibility: every step is documented so a third party can rebuild the lab from the accompanying repository.

Keywords: SIEM, Security Operations Centre, Wazuh, threat detection, endpoint telemetry, log analysis, home lab.

Contents

Abstract	1
1 Introduction	3
1.1 Aim and Objectives	3
1.2 Research Questions	3
1.3 Contributions	3
1.4 Report Structure	4
2 Literature Review and Background Knowledge	4
2.1 Overview	4
2.2 Security Operations Centres	4
2.3 Security Information and Event Management	5
2.4 Log Management and Endpoint Telemetry	5
2.5 Open-Source SIEM Platforms	5
2.6 Threat Detection Approaches and the MITRE ATT&CK Framework	6
2.7 Virtualised Security Labs and the Skills Gap	6
2.8 Related Works	7
3 Methodology	7
4 Implementation	7
5 Evaluation	7
6 Discussion	8
7 Conclusions	8
References	9

1 Introduction

Modern organisations generate enormous volumes of security-relevant data: authentication events, process executions, network connections, file changes, and system errors. Making sense of this data — and acting on it quickly enough to matter — is the work of a Security Operations Centre (SOC). At the centre of every SOC sits a Security Information and Event Management (SIEM) platform, which collects logs from across an estate, correlates them, and raises alerts an analyst can investigate.

For someone entering the cybersecurity profession, this creates a practical difficulty. Almost every junior SOC and security-analyst role expects familiarity with a SIEM, yet undergraduate study tends to teach the theory of detection and incident response without sustained hands-on exposure to the tools themselves. A graduate can describe what a SIEM does without ever having operated one.

This project was undertaken to close that gap directly. It designs, builds, and documents a working SOC home lab using only free and open-source software, producing both a functional environment and a reproducible record of how it was created.

1.1 Aim and Objectives

The **aim** of this project is to design, deploy, and operate a functional SOC home lab built on an open-source SIEM, capable of collecting and analysing security telemetry from multiple endpoints and supporting a realistic analyst triage workflow.

The aim is met through the following **objectives**:

- O1.** Deploy an open-source SIEM server on an isolated virtual network.
- O2.** Configure two monitored endpoints — a Windows host and a Linux host — each reporting to the SIEM.
- O3.** Enable enhanced host telemetry beyond default operating-system logs.
- O4.** Verify centralised log collection and establish a baseline of normal activity.
- O5.** Simulate a set of security events and observe the resulting alerts.
- O6.** Triage the alerts, classifying them and documenting the reasoning.
- O7.** Produce a fully reproducible record of the build so the lab can be rebuilt by a third party.

1.2 Research Questions

The project is framed around three research questions, revisited in the Discussion and Conclusions:

- **RQ1.** Can a credible SOC monitoring environment, capable of collecting endpoint telemetry and generating meaningful alerts, be built using only free and open-source tools?
- **RQ2.** What additional configuration is required beyond a default SIEM and operating-system installation to achieve detection-grade visibility?
- **RQ3.** To what extent does a single-machine virtualised lab realistically represent the work of a professional SOC, and where are the limits of that representation?

1.3 Contributions

This project contributes:

- A complete, working SOC home-lab build using Wazuh, deployed on an isolated network with Windows and Linux endpoints.
- A fully documented, step-by-step record enabling independent reproduction of the environment.
- A set of attack simulations paired with the alerts they produce and an analyst-style triage of each.
- An honest assessment of how closely such a lab represents real SOC work, including its limitations.

1.4 Report Structure

The remainder of this report is organised as follows. **Section 2** reviews the background literature on SOC's, SIEM technology, logging, and security home labs, and positions this work against related projects. **Section 3** sets out the general methodology and design decisions. **Section 4** documents the specific implementation of the lab. **Section 5** evaluates whether the lab meets its objectives. **Section 6** discusses challenges and revisits the research questions, and **Section 7** concludes and identifies future work.

2 Literature Review and Background Knowledge

2.1 Overview

This section establishes the background needed to understand the project. It begins with the function of a Security Operations Centre and the role of the analyst, then examines the technology that underpins it: Security Information and Event Management (SIEM) platforms, log management, and endpoint telemetry. It compares open-source and commercial SIEM options, outlines the main approaches to threat detection and the MITRE ATT&CK framework, and considers the use of virtualised home labs for developing practical skills. The section closes with a review of related work and a statement of how this project differs from it.

2.2 Security Operations Centres

A Security Operations Centre (SOC) is the organisational function responsible for continuously monitoring an environment, detecting malicious or anomalous activity, and coordinating the response to security incidents. Rather than a single technology, a SOC is best understood as the combination of people, process, and tooling working together to maintain an organisation's security posture.

SOC analysts are commonly organised into tiers. Tier 1 analysts perform the initial triage of incoming alerts, deciding quickly whether an alert represents genuine malicious activity (a true positive) or harmless behaviour that merely resembles it (a false positive). Alerts that cannot be dismissed are escalated to Tier 2 analysts for deeper investigation, and the most serious or complex cases reach Tier 3 staff who carry out incident response and proactive threat hunting. This triage-first model means that the largest share of day-to-day SOC work consists of rapidly reviewing, classifying, and documenting alerts.

The structured handling of incidents that a SOC supports is described in NIST's incident-handling guidance, which frames incident response as a lifecycle of preparation; detection and analysis; containment, eradication and recovery; and post-incident activity [2]. Detection and analysis — the stage at which raw data becomes an actionable alert — is the stage this project is most concerned with, and it depends almost entirely on the SOC's central tooling: the SIEM.

2.3 Security Information and Event Management

A SIEM platform is the technological core of a SOC. The term combines two earlier categories: Security Information Management (SIM), concerned with the long-term collection, storage, and analysis of log data, and Security Event Management (SEM), concerned with the real-time monitoring and correlation of events. A modern SIEM unifies both.

In practice, a SIEM performs several core functions. It *collects* log and event data from many different sources across an estate. It *normalises* that data, parsing records written in many different formats into a consistent structure so they can be compared and searched. It *correlates* events, applying rules that recognise patterns spanning multiple log sources — for example, linking a series of failed logins followed by a success into a single suspicious sequence. It *alerts* analysts when activity matches a rule of concern, and it provides *dashboards and search* so that analysts can investigate. Finally, it *retains* data for the periods required by forensic investigation and regulatory compliance.

SIEM platforms have evolved considerably. Early systems were primarily log aggregators; later generations added correlation and alerting; current platforms increasingly incorporate user and entity behaviour analytics, threat-intelligence enrichment, and automated response. Throughout this evolution, the underlying premise has remained constant: an organisation cannot detect what it does not collect and analyse, and the SIEM is the instrument through which that collection and analysis happens.

2.4 Log Management and Endpoint Telemetry

Because detection ultimately rests on log data, the quality of that data is decisive. NIST's guidance on computer security log management emphasises that logs are essential not only for detecting attacks but also for understanding them after the fact, and that effective log management requires deliberate planning of what to collect and how to handle it [3]. National guidance makes a similar point: logging must be designed with security outcomes in mind rather than left to operating-system defaults [6].

This distinction — between default logging and security-grade logging — is central to the present project. On Windows, the standard Event Log captures useful records such as logon events and service changes, but it does not by itself provide the fine-grained visibility that behavioural detection requires. **Sysmon** (System Monitor), a free tool from Microsoft's Sysinternals suite, addresses this by logging detailed information about process creation, network connections, image loads, file-creation timestamps, and related activity, written to a dedicated event channel [5]. This additional telemetry is what allows a SIEM to reason about *behaviour* rather than merely about discrete system messages.

Linux provides an equivalent capability through the Linux Audit framework and its userspace daemon, **auditd**. By defining audit rules, an administrator can record security-relevant system activity — notably the execution of commands through the **execve** system call — at the kernel level. As with Sysmon on Windows, this elevates the endpoint from producing general operational logs to producing detection-grade telemetry.

The recurring principle across both platforms is that endpoint visibility is something that must be deliberately engineered. A SIEM connected only to default logs will see far less than one fed by purpose-configured endpoint telemetry.

2.5 Open-Source SIEM Platforms

SIEM capability is available both as commercial products and as open-source software. Commercial platforms such as Splunk, Microsoft Sentinel, and IBM QRadar are powerful and widely

deployed, but they are typically licensed by data volume, which can make them costly — and largely inaccessible for independent learning and experimentation.

Several capable open-source alternatives exist, including the Elastic Stack (often deployed as Elastic Security), Security Onion, Graylog, and Wazuh. This project uses **Wazuh**. Wazuh originated as a fork of **OSSEC**, a long-established open-source host-based intrusion detection system [7], and has since grown into a full SIEM and extended detection and response (XDR) platform [12]. A Wazuh deployment consists of a central *manager* that analyses incoming data and applies detection rules, an *indexer* that stores and searches events, a *dashboard* that presents them, and lightweight *agents* installed on monitored endpoints. Beyond log analysis and alerting, Wazuh provides file-integrity monitoring, configuration assessment, vulnerability detection, and the mapping of alerts to the MITRE ATT&CK framework.

For the purpose of building practical skills, an open-source platform offers clear advantages. It can be deployed at no cost, it can be examined and reconfigured freely, and — importantly — the concepts it teaches, such as agent-based collection, log parsing, rule-based correlation, and alert triage, transfer directly to commercial platforms. The specific product differs; the underlying discipline does not.

2.6 Threat Detection Approaches and the MITRE ATT&CK Framework

SIEM platforms detect threats using two broad approaches. **Signature- or rule-based** detection matches activity against known patterns of malicious behaviour; it is precise and explainable, but it can only catch what has been defined in advance. **Anomaly-based** detection flags deviations from an established baseline of normal behaviour; it can in principle catch novel activity, but it is prone to raising alerts on benign deviations. NIST’s guidance on intrusion detection and prevention systems describes both approaches and notes that practical deployments generally combine them, since each compensates for the other’s weakness [9].

A persistent operational problem for SOCs is *alert fatigue*: when detection produces a high volume of low-value or false-positive alerts, analysts become desensitised, and genuine threats risk being overlooked among the noise. This makes the *tuning* of detection — raising true positives while suppressing false positives — as important as detection itself.

To make detection systematic, the security community has increasingly adopted the **MITRE ATT&CK** framework: a structured, continually updated knowledge base of adversary tactics and techniques drawn from real-world observations of attacks [10]. ATT&CK gives defenders a common vocabulary and a way to measure detection *coverage* — mapping which adversary techniques an environment can and cannot detect. While the detailed application of ATT&CK belongs to the planned follow-on project on detection engineering, it is introduced here because it is the framework against which modern detection capability is now routinely assessed.

2.7 Virtualised Security Labs and the Skills Gap

The cybersecurity profession is widely recognised to face a shortage of skilled practitioners, and industry reporting consistently identifies a mismatch between the volume of security-relevant activity organisations must handle and the workforce available to handle it [11]. For new entrants specifically, the gap is often less about theoretical knowledge than about practical, tool-level experience.

Virtualised home labs are a well-established response to this problem. Using a hypervisor such as VirtualBox or VMware, a learner can build a self-contained environment of multiple virtual machines on a single physical computer, at no cost and in complete isolation from any production network. Such a lab can be broken, rebuilt, and experimented on freely. For defensive security in particular, a home lab allows a learner to stand up a SIEM, generate security events safely,

and practise the monitoring and triage workflow that a SOC role demands — experience that is otherwise difficult to obtain before employment. Frameworks such as the CIS Critical Security Controls reinforce the value of this hands-on focus by expressing good security practice in terms of concrete, implementable actions rather than abstract principles [1].

2.8 Related Works

Building security home labs is an established practice, and several reference projects exist. The most prominent is **DetectionLab** by Chris Long [4], an automated project that provisions a pre-built Windows domain instrumented with logging and monitoring tooling. DetectionLab is comprehensive and highly automated; its strength — one-command provisioning — is also a limitation for a learner, as the automation abstracts away the very build steps from which much of the understanding comes. A large body of community material, including vendor tutorials and blog write-ups, similarly documents Wazuh and Elastic-based lab builds, though such material varies in depth and is rarely presented as a structured, reproducible whole.

This project is positioned deliberately against that backdrop, and differs in three respects. First, **the build is performed and documented manually and in full**: reproducibility is treated as a primary deliverable, with each step recorded so that the understanding — not only the result — is transferable. Second, **the evaluation is explicitly honest about realism**: rather than presenting a home lab as equivalent to a production SOC, the report sets out clearly where a single-machine virtualised environment does and does not represent real operations. Third, **the project is designed as the first stage of a two-part progression**: it establishes the monitoring foundation on which a planned second project, focused on detection engineering with MITRE ATT&CK and Atomic Red Team [8], will build. The contribution is therefore not a novel tool, but a clear, honest, and reproducible account of how a credible SOC monitoring environment is constructed and operated using open-source software.

3 Methodology

This section is to be written. It will describe the general approach taken to the project — the choice of an isolated virtualised environment, the selection of an open-source SIEM, the decision to document every step for reproducibility, and the iterative build-test-document method — at a general level, leaving specific commands and configuration to the Implementation section.

4 Implementation

This section is to be written. It will document the specific build of the lab: the isolated NAT network, the deployment of the Wazuh server, the configuration of the Windows and Linux endpoints with their agents and enhanced telemetry (Sysmon and auditd), and the attack simulations carried out. The step-by-step build records in the project repository provide the source material for this section.

5 Evaluation

This section is to be written. It will evaluate the lab against the objectives set in Section 1.1 — confirming that endpoints report to the SIEM, that simulated security events generate the expected alerts, and that each alert can be triaged — supported by evidence and metrics drawn from the running environment.

6 Discussion

This section is to be written. It will discuss the project as a whole, the challenges encountered during the build, and the research questions set out in Section 1.2 — in particular the extent to which a single-machine virtualised lab realistically represents professional SOC work.

7 Conclusions

This section is to be written. It will conclude the report, summarising what the project achieved against its aim and objectives, and identifying future work — including the planned second project on detection engineering with MITRE ATT&CK.

References

- [1] Center for Internet Security. Cis critical security controls, version 8. <https://www.cisecurity.org/controls>, 2021. Accessed: 2026.
- [2] Paul Cichonski, Tom Millar, Tim Grance, and Karen Scarfone. Computer security incident handling guide (sp 800-61 revision 2). Nist special publication, National Institute of Standards and Technology (NIST), 2012.
- [3] Karen Kent and Murugiah Souppaya. Guide to computer security log management (sp 800-92). Nist special publication, National Institute of Standards and Technology (NIST), 2006.
- [4] Chris Long. Detectionlab. <https://github.com/clong/DetectionLab>, 2023. Accessed: 2026.
- [5] Microsoft Corporation. Sysmon — windows sysinternals. <https://learn.microsoft.com/sysinternals/downloads/sysmon>, 2025. Accessed: 2026.
- [6] National Cyber Security Centre (NCSC). Introduction to logging for security purposes. <https://www.ncsc.gov.uk/guidance/introduction-logging-security-purposes>, 2025. Accessed: 2026.
- [7] OSSEC Project. Ossec — open source hids. <https://www.ossec.net>, 2025. Accessed: 2026.
- [8] Red Canary. Atomic red team. <https://github.com/redcanaryco/atomic-red-team>, 2025. Accessed: 2026.
- [9] Karen Scarfone and Peter Mell. Guide to intrusion detection and prevention systems (idps) (sp 800-94). Nist special publication, National Institute of Standards and Technology (NIST), 2007.
- [10] Blake E. Strom, Andy Applebaum, Doug P. Miller, Kathryn C. Nickels, Adam G. Pennington, and Cody B. Thomas. Mitre att&ck: Design and philosophy. Technical report, The MITRE Corporation, 2020.
- [11] Verizon. Data breach investigations report (dbir). <https://www.verizon.com/business/resources/reports/dbir/>, 2024. Accessed: 2026.
- [12] Wazuh Inc. Wazuh documentation. <https://documentation.wazuh.com>, 2025. Accessed: 2026.